2b_Scenario1_Diabetes_Starter.R

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2026-10-26

```
################################################################################
# FINAL PROJECT SCENARIO 1: UNDERSTANDING DIABETES PROGRESSION
# Starter R script
#
# This script is designed to run from top to bottom.
# It gives you working code for the required analysis, together with many
# comments explaining what each section does.
#
# IMPORTANT
# 1. The code produces results, but your task is to interpret them.
# 2. Do not copy raw R output into your report without explanation.
# 3. Do not treat an association as evidence of causation.
# 4. You may adapt the code and choose different predictors when you can
#    justify your choices.
################################################################################
```

```
# 0. SETUP
```

```
# The project uses a small number of common packages.
# We use requireNamespace() to check whether each package is available.
# If a package is missing, R will install it before loading it.
needed_packages <- c("ggplot2", "dplyr", "broom")

for (pkg in needed_packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg)
  }
}

library(ggplot2)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.6.1
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.6.1
```

```r
# Make printed numbers a little easier to read.
options(digits = 4)


# ============================================================================
# QUESTION 1. UNDERSTAND AND PREPARE THE DATA
# ============================================================================

# A local copy of the dataset is supplied with the project.
# Keep diabetes.tab.txt in the same folder as this R file.
# The original public source is:
# https://www4.stat.ncsu.edu/~boos/var.select/diabetes.tab.txt
data_file <- "diabetes.tab.txt"

# If R cannot find the file in the current working directory, choose it manually.
# This avoids making the graded project depend on an external website being online.
if (!file.exists(data_file)) {
  message("diabetes.tab.txt was not found in the current working directory.")
  message("Please choose the supplied diabetes.tab.txt file.")
  data_file <- file.choose()
}

# Read the supplied tab-separated file.
# header = TRUE means that the first row contains the column names.
diabetes <- read.table(data_file, header = TRUE, sep = "\t")

# Convert column names to lower case so they are easier to type consistently.
names(diabetes) <- tolower(names(diabetes))

# The source dataset uses 'y' as the disease-progression outcome.
# Rename it to something clearer for the rest of the analysis.
if ("y" %in% names(diabetes)) {
  diabetes <- diabetes %>% rename(progression = y)
}

# Look at the first few rows.
head(diabetes)
```

```
##   age sex  bmi  bp  s1    s2 s3 s4    s5 s6 progression
## 1  59   2 32.1 101 157  93.2 38  4 4.860 87         151
## 2  48   1 21.6  87 183 103.2 70  3 3.892 69          75
## 3  72   2 30.5  93 156  93.6 41  4 4.673 85         141
## 4  24   1 25.3  84 198 131.4 40  5 4.890 89         206
## 5  50   1 23.0 101 192 125.4 52  4 4.290 80         135
## 6  23   1 22.6  89 139  64.8 61  2 4.190 68          97
```

```r
# Number of rows and columns.
dim(diabetes)
```

```
## [1] 442  11
```

```
# Variable names.
names(diabetes)
```

```
##  [1] "age"         "sex"         "bmi"         "bp"          "s1"
##  [6] "s2"          "s3"          "s4"          "s5"          "s6"
## [11] "progression"
```

```
# Structure and data types.
str(diabetes)
```

```
## 'data.frame':    442 obs. of  11 variables:
##  $ age        : num  59 48 72 24 50 23 36 66 60 29 ...
##  $ sex        : num  2 1 2 1 1 1 2 2 2 1 ...
##  $ bmi        : num  32.1 21.6 30.5 25.3 23 22.6 22 26.2 32.1 30 ...
##  $ bp         : num  101 87 93 84 101 89 90 114 83 85 ...
##  $ s1         : num  157 183 156 198 192 139 160 255 179 180 ...
##  $ s2         : num  93.2 103.2 93.6 131.4 125.4 ...
##  $ s3         : num  38 70 41 40 52 61 50 56 42 43 ...
##  $ s4         : num  4 3 4 5 4 2 3 4.55 4 4 ...
##  $ s5         : num  4.86 3.89 4.67 4.89 4.29 ...
##  $ s6         : num  87 69 85 89 80 68 82 92 94 88 ...
##  $ progression: num  151 75 141 206 135 97 138 63 110 310 ...
```

```
# The variable 'sex' is numerically coded in the source data, but it represents
# categories rather than a continuous measurement. Check the values present,
# then convert it to a factor so R treats it as categorical in later analyses.
unique(diabetes$sex)
```

```
## [1] 2 1
```

```
diabetes$sex <- factor(diabetes$sex)

# Count missing values in each variable.
colSums(is.na(diabetes))
```

```
##         age         sex         bmi          bp          s1          s2
##           0           0           0           0           0           0
##          s3          s4          s5          s6 progression
##           0           0           0           0           0
```

```
# Total number of missing values.
sum(is.na(diabetes))
```

```
## [1] 0
```

```
# Check for duplicated rows.
sum(duplicated(diabetes))
```

```
## [1] 0
```

```
# Basic summary of every variable.
summary(diabetes)
```

```
##       age       sex          bmi              bp              s1
##  Min.   :19.0   1:235   Min.   :18.0   Min.   : 62.0   Min.   : 97
##  1st Qu.:38.2   2:207   1st Qu.:23.2   1st Qu.: 84.0   1st Qu.:164
##  Median :50.0           Median :25.7   Median : 93.0   Median :186
##  Mean   :48.5           Mean   :26.4   Mean   : 94.6   Mean   :189
##  3rd Qu.:59.0           3rd Qu.:29.3   3rd Qu.:105.0   3rd Qu.:210
##  Max.   :79.0           Max.   :42.2   Max.   :133.0   Max.   :301
##        s2              s3             s4             s5             s6
##  Min.   : 41.6   Min.   :22.0   Min.   :2.00   Min.   :3.26   Min.   : 58.0
##  1st Qu.: 96.0   1st Qu.:40.2   1st Qu.:3.00   1st Qu.:4.28   1st Qu.: 83.2
##  Median :113.0   Median :48.0   Median :4.00   Median :4.62   Median : 91.0
##  Mean   :115.4   Mean   :49.8   Mean   :4.07   Mean   :4.64   Mean   : 91.3
##  3rd Qu.:134.5   3rd Qu.:57.8   3rd Qu.:5.00   3rd Qu.:5.00   3rd Qu.: 98.0
##  Max.   :242.4   Max.   :99.0   Max.   :9.09   Max.   :6.11   Max.   :124.0
##   progression
##  Min.   : 25
##  1st Qu.: 87
##  Median :140
##  Mean   :152
##  3rd Qu.:212
##  Max.   :346
```

```
# FOR YOUR REPORT
# - What does one row represent? One row represents one observation of 442 patients
# - Which variable is the outcome? The variable that is the outcome is the Y
# value that was named after as 'progression'
# - Which variables are predictors?
# - Which variables are numerical, and which are categorical in meaning?
# - Are there missing values, duplicated rows or obviously problematic values?
# - Why do health-related data require careful and responsible interpretation?


# ============================================================================
# QUESTION 2. EXPLORE THE DATA
# ============================================================================

# A helper function for common descriptive statistics.
describe_numeric <- function(x) {
  c(
    n = sum(!is.na(x)),
    mean = mean(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
    sd = sd(x, na.rm = TRUE),
    min = min(x, na.rm = TRUE),
    q1 = quantile(x, 0.25, na.rm = TRUE),
    q3 = quantile(x, 0.75, na.rm = TRUE),
    max = max(x, na.rm = TRUE),
    iqr = IQR(x, na.rm = TRUE)
  )
}

# Descriptive statistics for the outcome.
describe_numeric(diabetes$progression)
```

```
##      n   mean median     sd    min q1.25% q3.75%    max    iqr
## 442.00 152.13 140.50  77.09  25.00  87.00 211.50 346.00 124.50
```

```
# Descriptive statistics for BMI.
describe_numeric(diabetes$bmi)
```

```
##       n    mean  median      sd     min  q1.25%  q3.75%     max     iqr
## 442.000  26.376  25.700   4.418  18.000  23.200  29.275  42.200   6.075
```

```
# Descriptive statistics for age.
describe_numeric(diabetes$age)
```

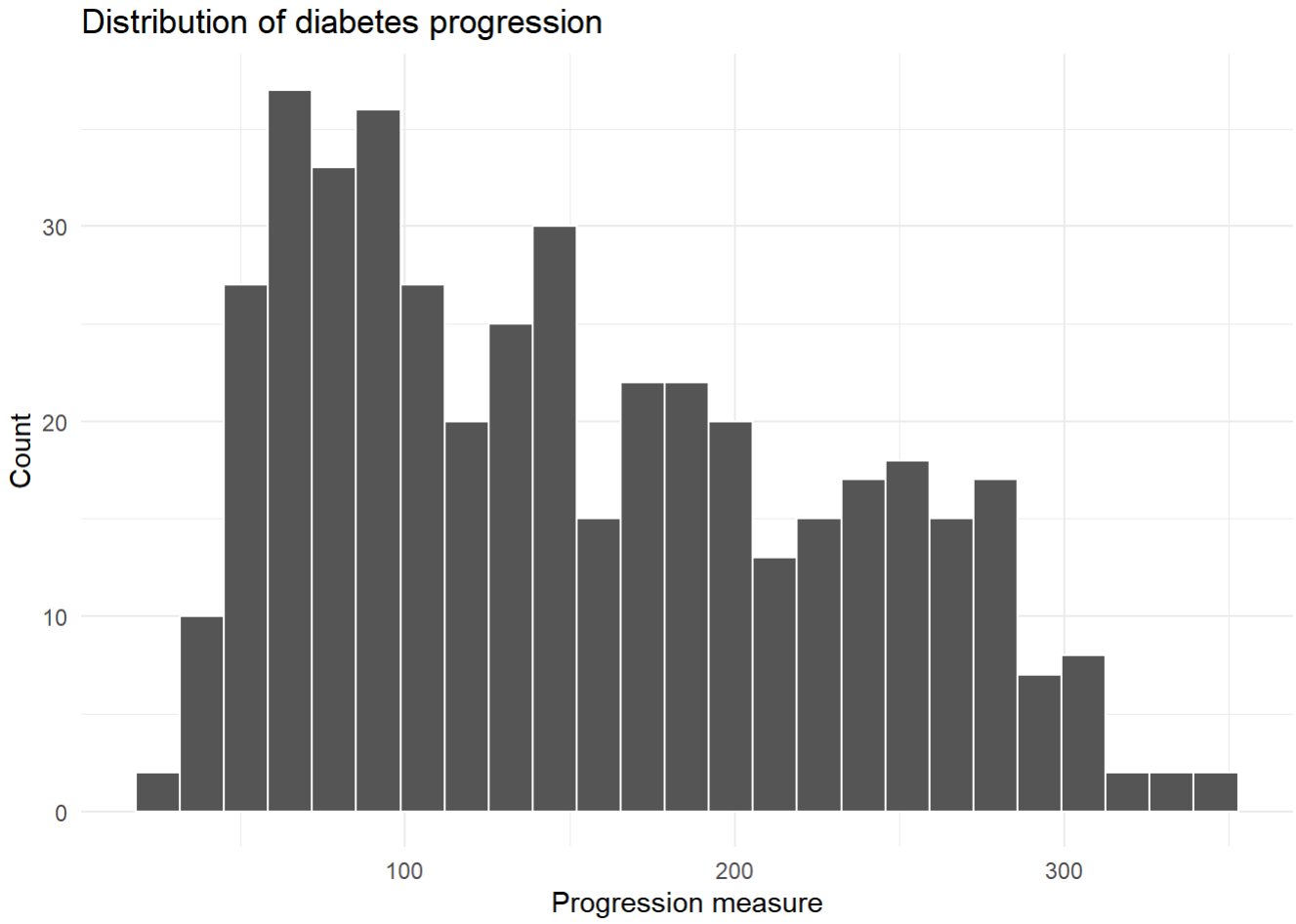
```
##      n   mean median     sd    min q1.25% q3.75%    max    iqr
## 442.00  48.52  50.00  13.11  19.00  38.25  59.00  79.00  20.75
```

```r
# A compact descriptive table for all numeric variables.
# You do not need to put the whole table in your report.
descriptive_table <- diabetes %>%
  summarise(across(
    where(is.numeric),
    list(
      mean = ~mean(.x, na.rm = TRUE),
      median = ~median(.x, na.rm = TRUE),
      sd = ~sd(.x, na.rm = TRUE),
      min = ~min(.x, na.rm = TRUE),
      max = ~max(.x, na.rm = TRUE)
    )
  ))

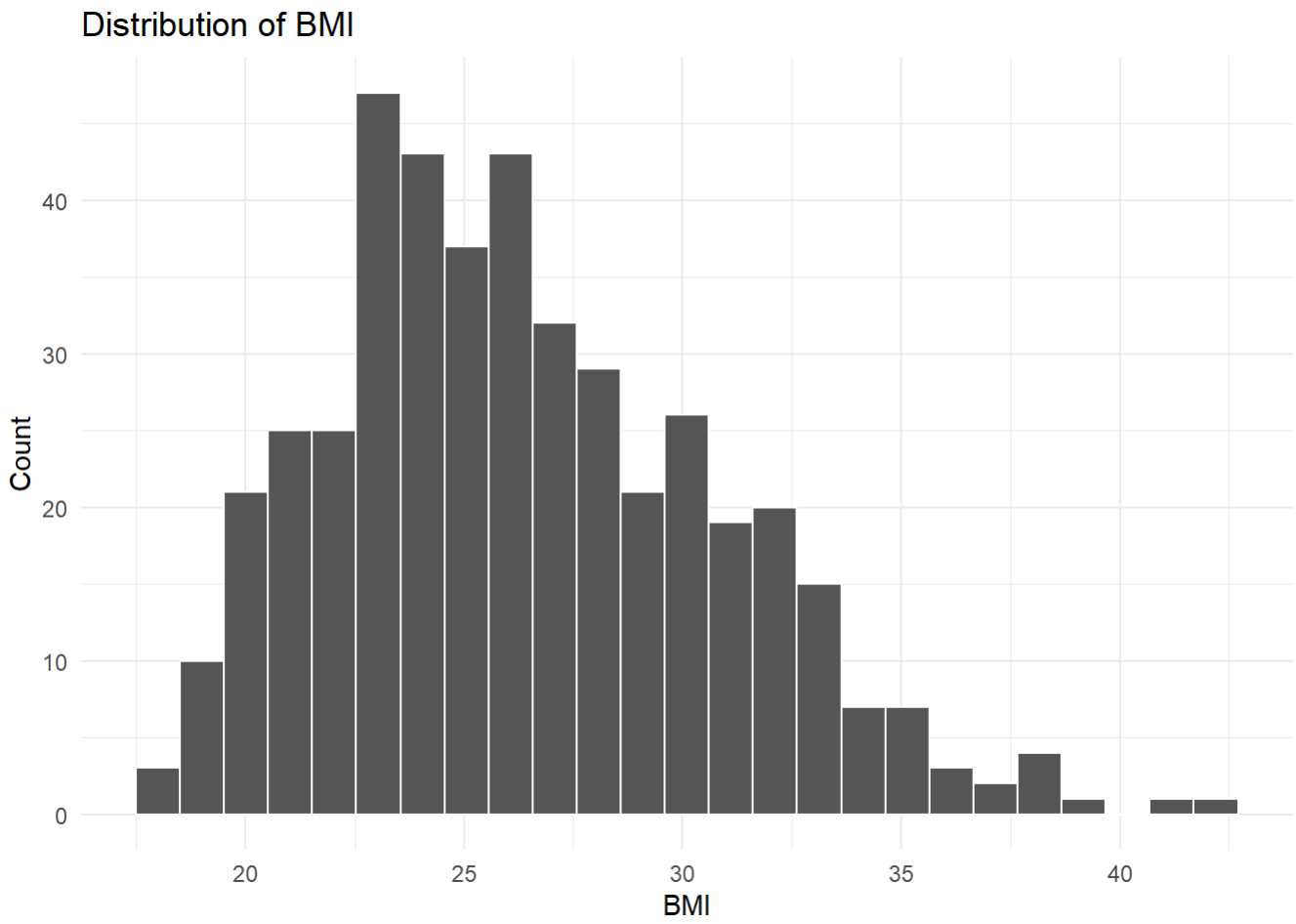
print(descriptive_table)
```

```
##   age_mean age_median age_sd age_min age_max bmi_mean bmi_median bmi_sd bmi_min
## 1    48.52         50  13.11      19      79    26.38       25.7  4.418      18
##   bmi_max bp_mean bp_median bp_sd bp_min bp_max s1_mean s1_median s1_sd s1_min
## 1    42.2   94.65        93 13.83     62    133   189.1       186 34.61     97
##   s1_max s2_mean s2_median s2_sd s2_min s2_max s3_mean s3_median s3_sd s3_min
## 1    301   115.4       113 30.41   41.6  242.4   49.79        48 12.93     22
##   s3_max s4_mean s4_median s4_sd s4_min s4_max s5_mean s5_median  s5_sd s5_min
## 1     99    4.07         4  1.29      2   9.09   4.641      4.62 0.5224  3.258
##   s5_max s6_mean s6_median s6_sd s6_min s6_max progression_mean
## 1  6.107   91.26        91  11.5     58    124            152.1
##   progression_median progression_sd progression_min progression_max
## 1              140.5          77.09              25             346
```

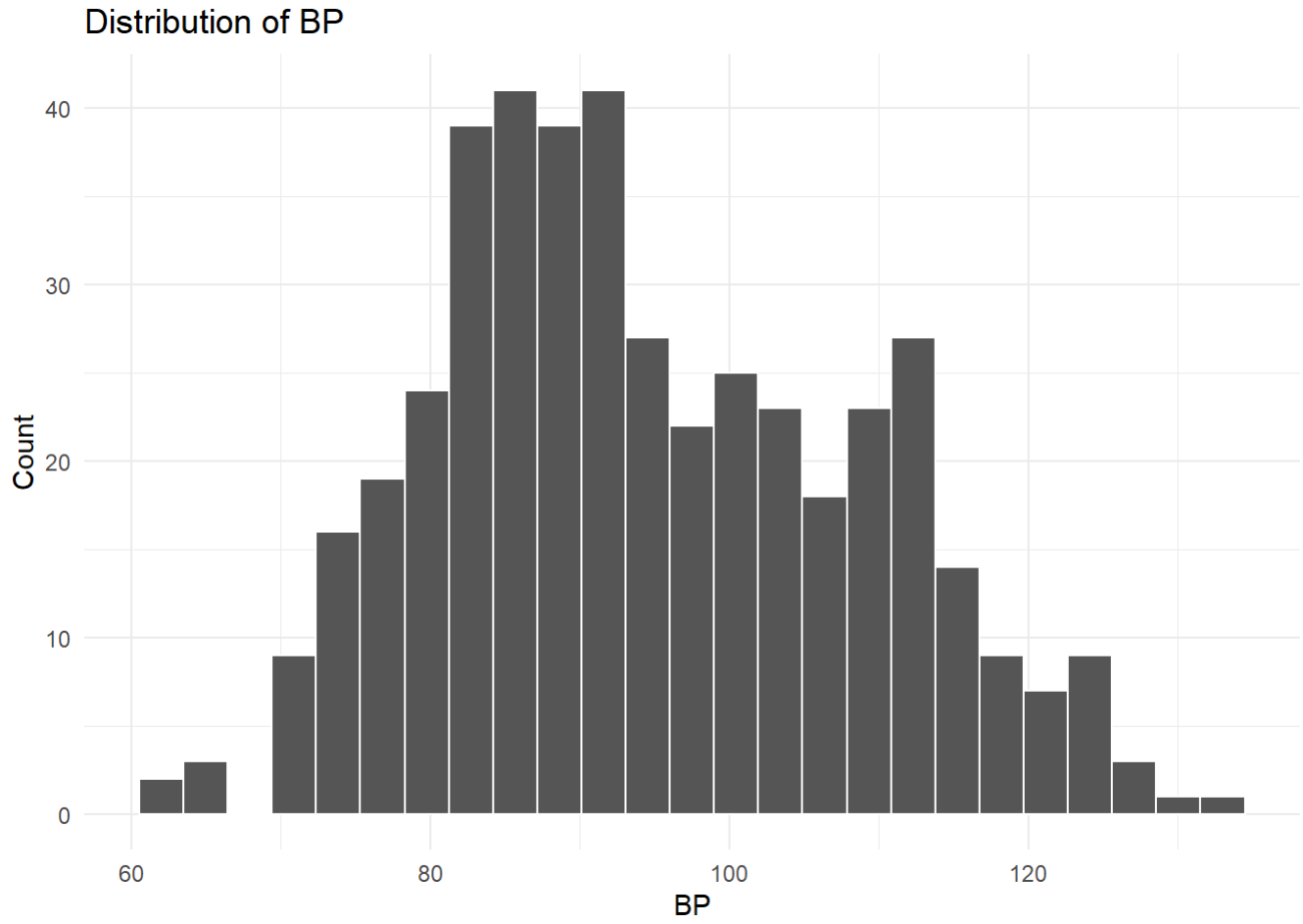
```r
# Distribution of diabetes progression.
ggplot(diabetes, aes(x = progression)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of diabetes progression",
    x = "Progression measure",
    y = "Count"
  ) +
  theme_minimal()
```



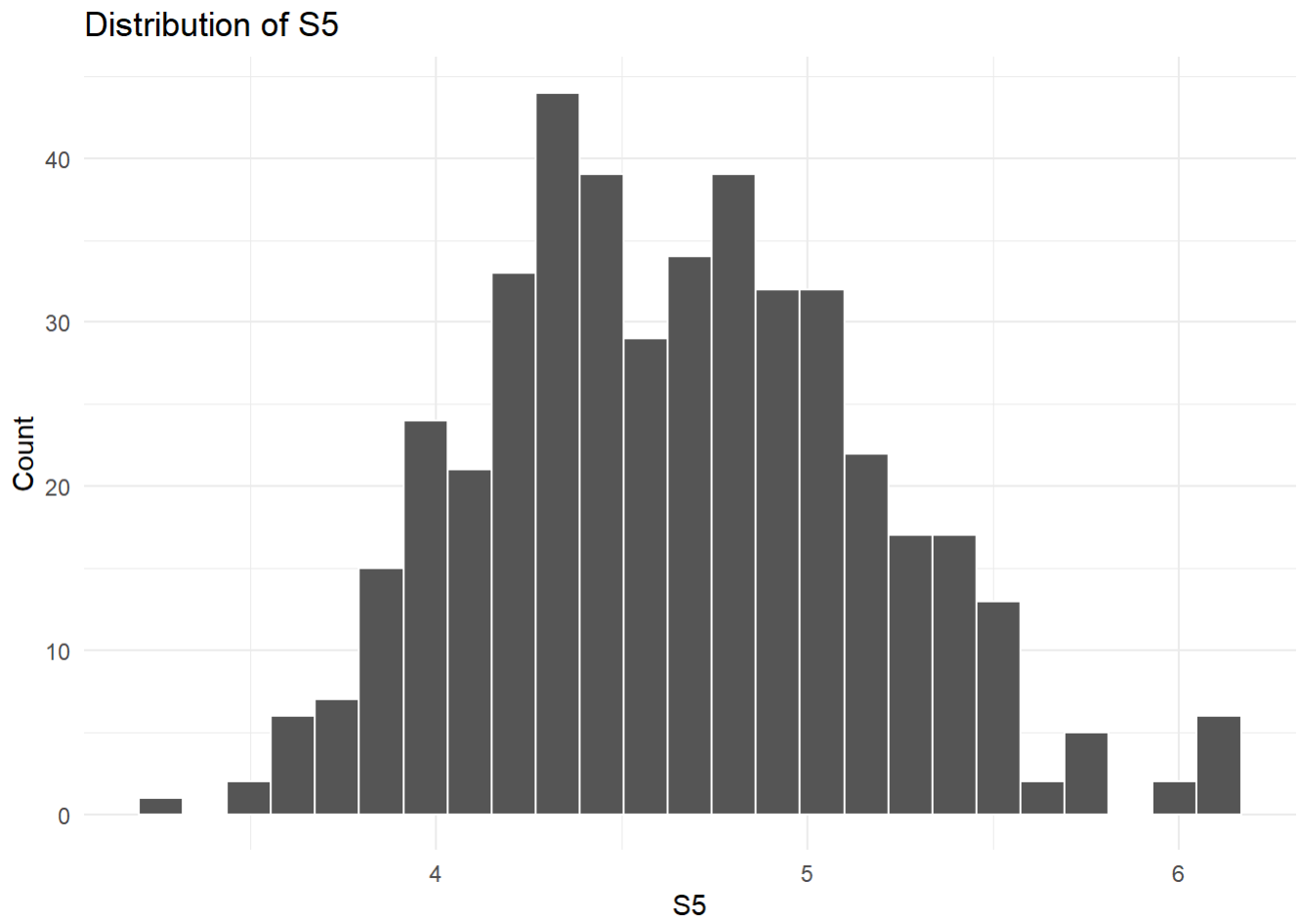

```
# Distribution of BMI.
ggplot(diabetes, aes(x = bmi)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of BMI",
    x = "BMI",
    y = "Count"
  ) +
  theme_minimal()
```



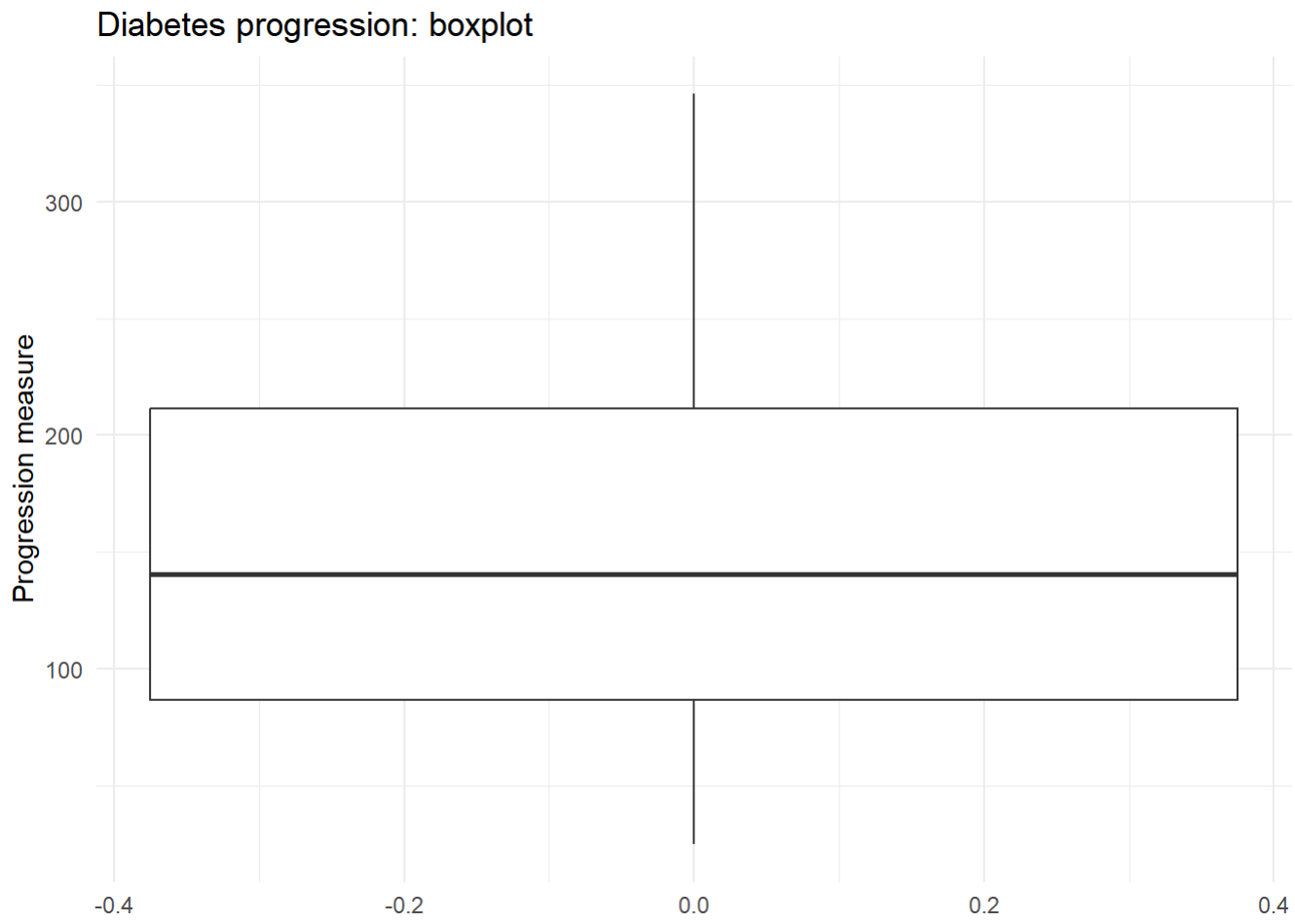

```
# Distribution of BP.
ggplot(diabetes, aes(x = bp)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of BP",
    x = "BP",
    y = "Count"
  ) +
  theme_minimal()
```




```
# Distribution of S5.
ggplot(diabetes, aes(x = s5)) +
  geom_histogram(bins = 25, colour = "white") +
  labs(
    title = "Distribution of S5",
    x = "S5",
    y = "Count"
  ) +
  theme_minimal()
```




```
# One additional useful visualisation: boxplot of progression.
ggplot(diabetes, aes(y = progression)) +
  geom_boxplot() +
  labs(
    title = "Diabetes progression: boxplot",
    y = "Progression measure",
    x = NULL
  ) +
  theme_minimal()
```




```
# You may replace or supplement the boxplot with another plot that you find
# more informative. For example, you could examine age, blood pressure or s5.

# FOR YOUR REPORT
# - What do the centre and spread of the main variables look like?
# - Are the distributions approximately symmetric or clearly skewed?
# - Are there unusual observations?
# - Which features of the data matter before modelling begins?


# ============================================================================
# QUESTION 3. INVESTIGATE RELATIONSHIPS
# ============================================================================

# Keep only genuinely numerical measurement variables for Pearson correlations.
# 'sex' has already been converted to a factor, so it is excluded automatically.
numeric_data <- diabetes %>% select(where(is.numeric))

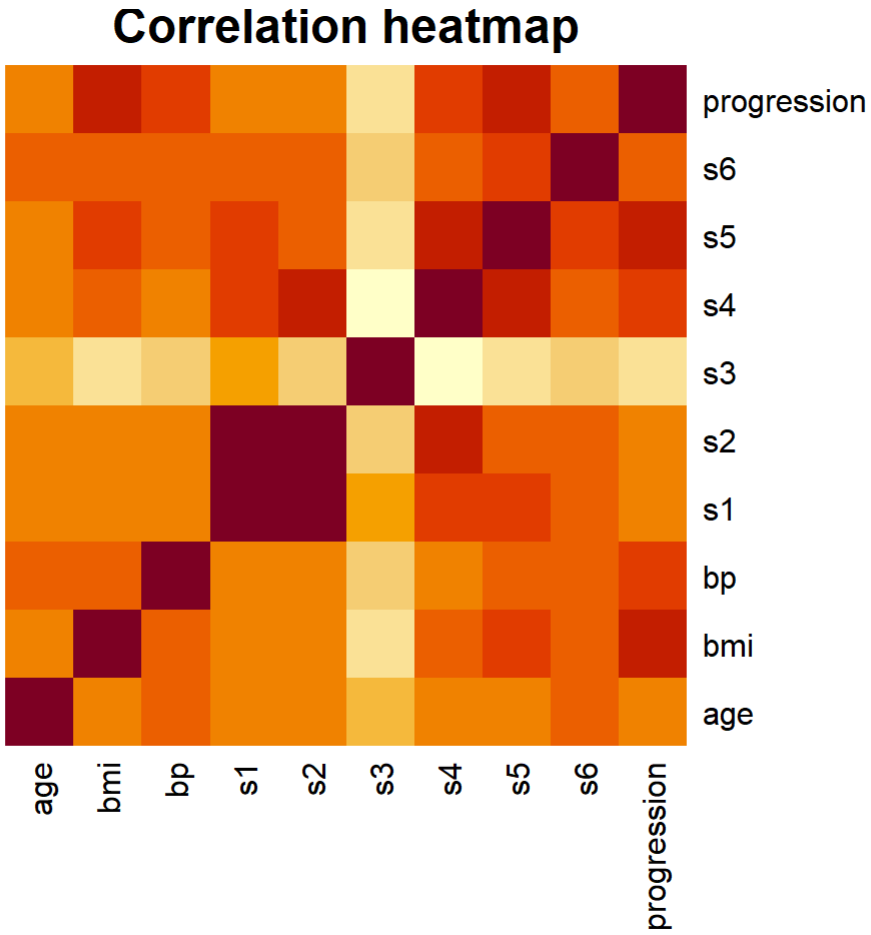
# Correlation matrix.
cor_matrix <- cor(numeric_data, use = "complete.obs", method = "pearson")
round(cor_matrix, 2)
```

```
##                age   bmi    bp   s1    s2    s3    s4    s5    s6 progression
## age          1.00  0.19  0.34 0.26  0.22 -0.08  0.20  0.27  0.30        0.19
## bmi          0.19  1.00  0.40 0.25  0.26 -0.37  0.41  0.45  0.39        0.59
## bp           0.34  0.40  1.00 0.24  0.19 -0.18  0.26  0.39  0.39        0.44
## s1           0.26  0.25  0.24 1.00  0.90  0.05  0.54  0.52  0.33        0.21
## s2           0.22  0.26  0.19 0.90  1.00 -0.20  0.66  0.32  0.29        0.17
## s3          -0.08 -0.37 -0.18 0.05 -0.20  1.00 -0.74 -0.40 -0.27       -0.39
## s4           0.20  0.41  0.26 0.54  0.66 -0.74  1.00  0.62  0.42        0.43
## s5           0.27  0.45  0.39 0.52  0.32 -0.40  0.62  1.00  0.46        0.57
## s6           0.30  0.39  0.39 0.33  0.29 -0.27  0.42  0.46  1.00        0.38
## progression  0.19  0.59  0.44 0.21  0.17 -0.39  0.43  0.57  0.38        1.00
```

```
# Correlations of all numeric variables with progression.
# The outcome itself will appear with correlation 1, so focus on the predictors.
progression_correlations <- sort(
  cor_matrix[, "progression"],
  decreasing = TRUE
)
print(progression_correlations)
```

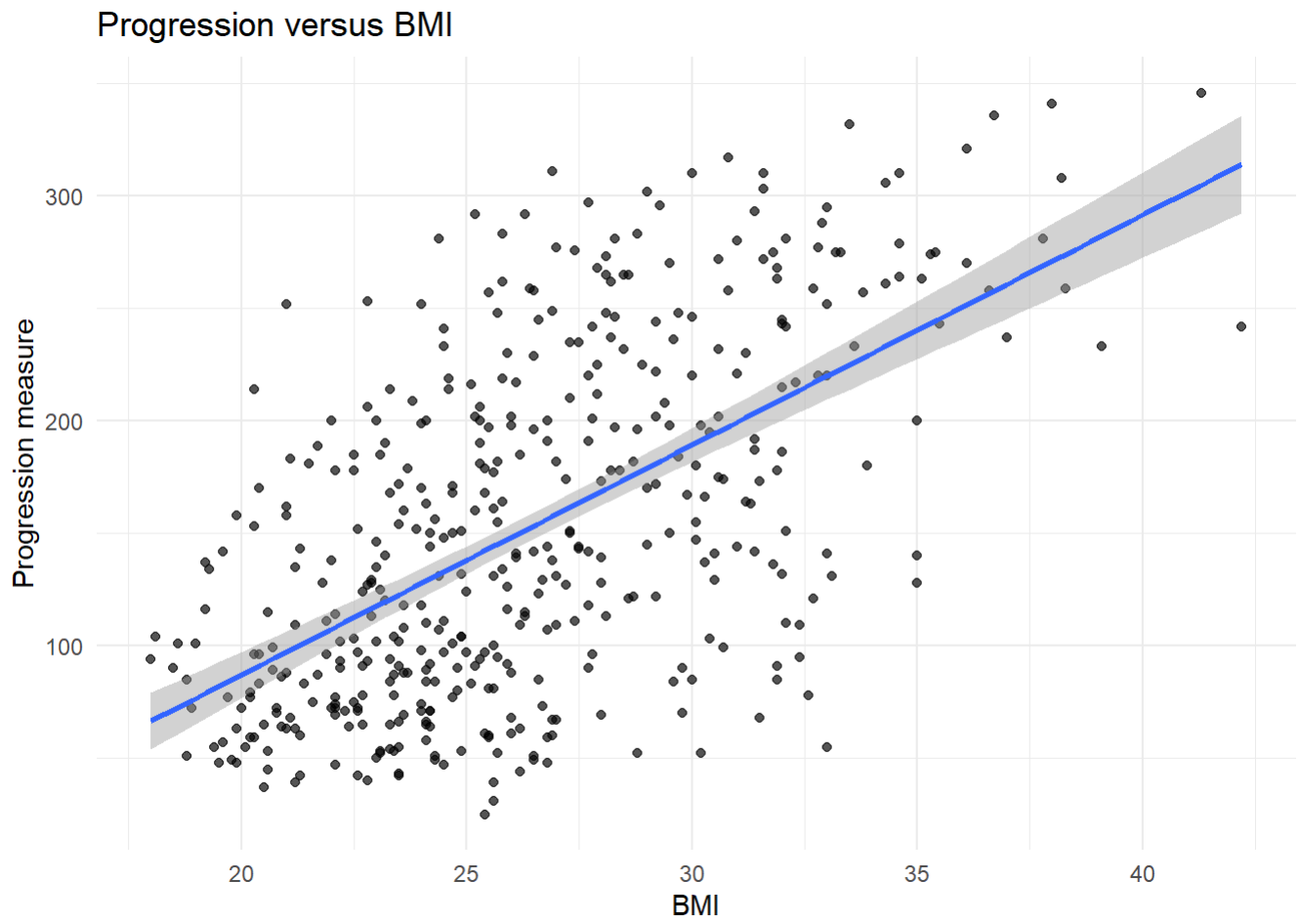
```
## progression         bmi          s5          bp          s4          s6
##      1.0000      0.5865      0.5659      0.4415      0.4305      0.3825
##          s1         age          s2          s3
##      0.2120      0.1879      0.1741     -0.3948
```

```
# Simple heatmap using base R.
# Rowv = NA and Colv = NA keep the variables in their original order instead
# of reordering them with clustering dendrograms.
heatmap(
  cor_matrix,
  Rowv = NA,
  Colv = NA,
  scale = "none",
  symm = TRUE,
  margins = c(7, 7),
  main = "Correlation heatmap"
)
```



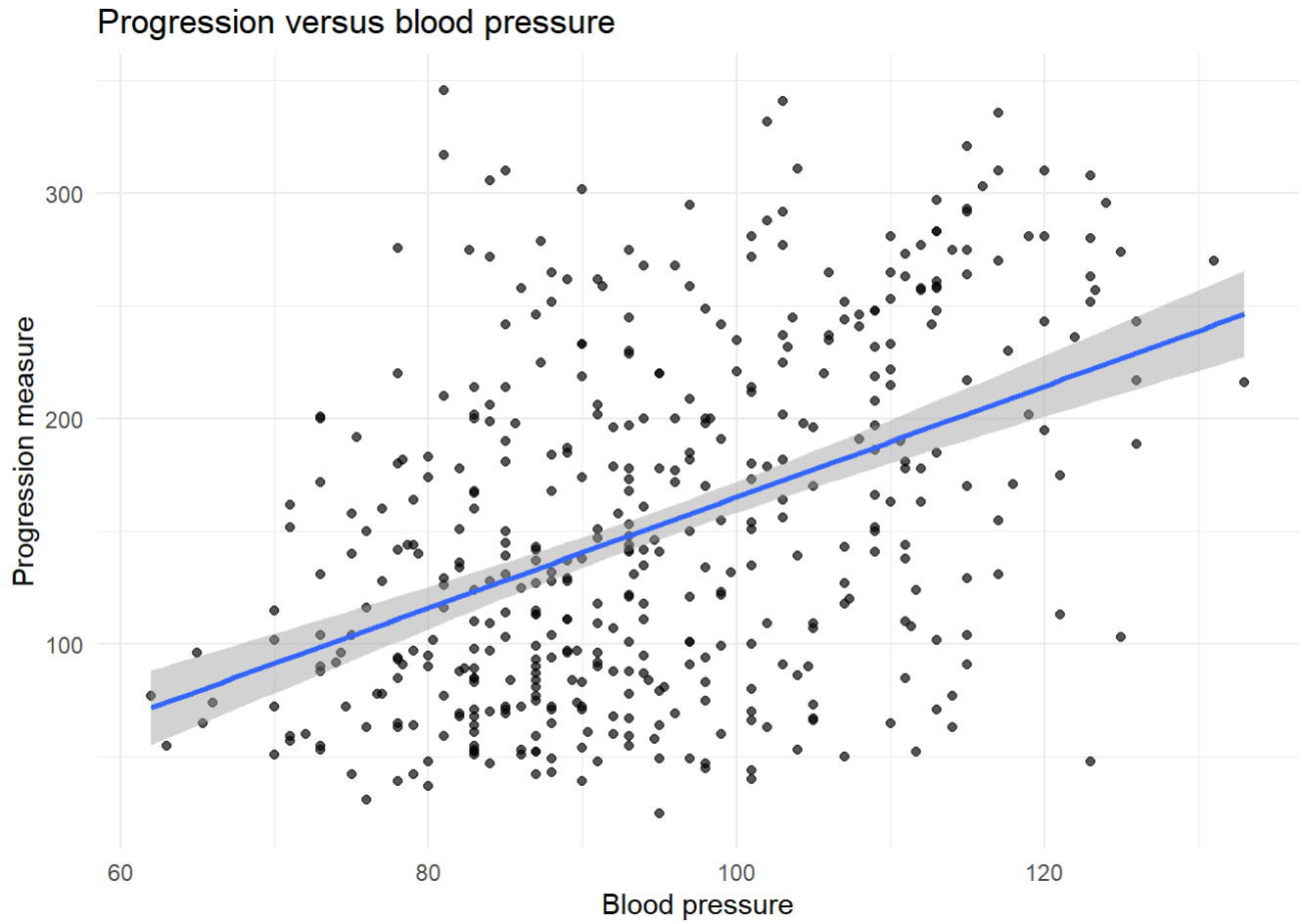

```
# Scatterplot 1: BMI and progression.
ggplot(diabetes, aes(x = bmi, y = progression)) +
  geom_point(alpha = 0.65) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Progression versus BMI",
    x = "BMI",
    y = "Progression measure"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Scatterplot 2: blood pressure and progression.
ggplot(diabetes, aes(x = bp, y = progression)) +
  geom_point(alpha = 0.65) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Progression versus blood pressure",
    x = "Blood pressure",
    y = "Progression measure"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# You may replace the second scatterplot with another relationship that looks
# more informative after examining the correlation matrix.

# FOR YOUR REPORT
# - Which predictors are most strongly associated with progression?
# - Which predictors are strongly related to each other?
# - Could some predictors carry overlapping information?
# - Which variables look most promising for regression analysis, and why?


# ============================================================================
# QUESTION 4. BUILD AND INTERPRET REGRESSION MODELS
# ============================================================================
```

```
# BMI is used as the starting predictor because it is meaningful and commonly
# examined in this dataset. You may choose another predictor if you justify it.
simple_model <- lm(progression ~ bmi, data = diabetes)

# Standard model summary.
summary(simple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -164.92  -43.57   -8.65   46.34  154.88
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -117.773     18.019   -6.54  1.8e-10 ***
## bmi           10.233      0.674   15.19  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 62.5 on 440 degrees of freedom
## Multiple R-squared:  0.344,  Adjusted R-squared:  0.342
## F-statistic:  231 on 1 and 440 DF,  p-value: <2e-16
```

```
# Tidy coefficient table, including confidence intervals.
tidy(simple_model, conf.int = TRUE)
```

```
## # A tibble: 2 × 7
##   term        estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept)   -118.     18.0       -6.54 1.75e-10  -153.       -82.4
## 2 bmi             10.2     0.674     15.2  3.47e-42     8.91      11.6
```

```
# Model-level statistics such as R-squared and adjusted R-squared.
glance(simple_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.344         0.342  62.5      231. 3.47e-42     1 -2454. 4914. 4926.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Coefficients for writing the fitted equation.
coef(simple_model)
```

```
## (Intercept)         bmi
##     -117.77       10.23
```

```
# FOR YOUR REPORT
# - Write the fitted equation.
# - Interpret the slope in context.
# - Does the intercept have a useful real-world interpretation?
# - What does R-squared mean here?
```

```r
# This is a ready-to-run starting model using three predictors.
# You may change the predictor set after examining your EDA and correlations.
multiple_model <- lm(
  progression ~ bmi + s5,
  data = diabetes
)

summary(multiple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi + s5, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -159.08  -42.75   -5.64   43.38  145.94
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -299.958     24.935  -12.03   <2e-16 ***
## bmi            7.276      0.684   10.64   <2e-16 ***
## s5            56.056      5.786    9.69   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 56.8 on 439 degrees of freedom
## Multiple R-squared:  0.459,  Adjusted R-squared:  0.457
## F-statistic:  187 on 2 and 439 DF,  p-value: <2e-16
```

```r
tidy(multiple_model, conf.int = TRUE)
```

```
## # A tibble: 3 × 7
##   term        estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept)  -300.      24.9      -12.0  5.25e-29  -349.     -251.
## 2 bmi             7.28     0.684     10.6  1.17e-23     5.93      8.62
## 3 s5             56.1      5.79       9.69 3.04e-20    44.7      67.4
```

```r
glance(multiple_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.459         0.457  56.8      187. 2.25e-59     2 -2411. 4830. 4847.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
multiple_model <- lm(
  progression ~ bmi + bp + s5,
  data = diabetes
)

summary(multiple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi + bp + s5, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -140.23  -40.64   -2.19   38.27  139.80
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -334.881     25.878  -12.94  < 2e-16 ***
## bmi            6.500      0.697    9.32  < 2e-16 ***
## bp             0.903      0.217    4.17  3.7e-05 ***
## s5            49.577      5.890    8.42  5.6e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 55.8 on 438 degrees of freedom
## Multiple R-squared:  0.48,   Adjusted R-squared:  0.477
## F-statistic:  135 on 3 and 438 DF,  p-value: <2e-16
```

```
tidy(multiple_model, conf.int = TRUE)
```

```
## # A tibble: 4 × 7
##   term        estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept) -335.       25.9      -12.9  1.16e-32 -386.      -284.
## 2 bmi            6.50      0.697      9.32 5.55e-19    5.13       7.87
## 3 bp             0.903     0.217      4.17 3.74e- 5    0.477      1.33
## 4 s5            49.6       5.89       8.42 5.56e-16   38.0       61.2
```

```
glance(multiple_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.480         0.477  55.8      135. 7.16e-62     3 -2403. 4815. 4836.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
multiple_model <- lm(
  progression ~ bmi + age + s5,
  data = diabetes
)

summary(multiple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi + age + s5, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -157.79  -42.70   -6.06   43.94  145.47
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -300.7034    25.1658  -11.95   <2e-16 ***
## bmi            7.2641     0.6868   10.58   <2e-16 ***
## age            0.0502     0.2152    0.23     0.82
## s5            55.7606     5.9297    9.40   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 56.9 on 438 degrees of freedom
## Multiple R-squared:  0.46,   Adjusted R-squared:  0.456
## F-statistic:  124 on 3 and 438 DF,  p-value: <2e-16
```

```
tidy(multiple_model, conf.int = TRUE)
```

```
## # A tibble: 4 × 7
##   term         estimate std.error statistic  p.value conf.low conf.high
##   <chr>           <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept) -301.        25.2     -11.9   1.12e-28 -350.     -251.
## 2 bmi            7.26       0.687    10.6   1.95e-23    5.91      8.61
## 3 age            0.0502     0.215     0.233 8.16e- 1   -0.373     0.473
## 4 s5            55.8        5.93      9.40  2.97e-19   44.1      67.4
```

```
glance(multiple_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.460         0.456  56.9      124. 3.38e-58     3 -2411. 4832. 4853.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
# Compare the simple and multiple models using measures taught in the course.
# R-squared shows explained variation, while adjusted R-squared takes model
# complexity into account when several predictors are used.
model_comparison <- bind_rows(
  glance(simple_model) %>% mutate(model = "Simple: BMI"),
  glance(multiple_model) %>% mutate(model = "Multiple: BMI + BP + S5")
) %>%
  select(model, r.squared, adj.r.squared, sigma)
multiple_model <- lm(
  progression ~ bmi + s5,
  data = diabetes
)

summary(multiple_model)
```

```
##
## Call:
## lm(formula = progression ~ bmi + s5, data = diabetes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -159.08  -42.75   -5.64   43.38  145.94
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -299.958     24.935  -12.03   <2e-16 ***
## bmi            7.276      0.684   10.64   <2e-16 ***
## s5            56.056      5.786    9.69   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 56.8 on 439 degrees of freedom
## Multiple R-squared:  0.459,  Adjusted R-squared:  0.457
## F-statistic:  187 on 2 and 439 DF,  p-value: <2e-16
```

```
tidy(multiple_model, conf.int = TRUE)
```

```
## # A tibble: 3 × 7
##   term        estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept)  -300.      24.9      -12.0  5.25e-29  -349.     -251.
## 2 bmi             7.28     0.684     10.6  1.17e-23     5.93      8.62
## 3 s5             56.1      5.79       9.69 3.04e-20    44.7      67.4
```

```
glance(multiple_model)
```

```
## # A tibble: 1 × 12
##   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
##       <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     0.459         0.457  56.8      187. 2.25e-59     2 -2411. 4830. 4847.
## # ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
# Compare the simple and multiple models using measures taught in the course.
# R-squared shows explained variation, while adjusted R-squared takes model
# complexity into account when several predictors are used.
model_comparison <- bind_rows(
  glance(simple_model) %>% mutate(model = "Simple: BMI"),
  glance(multiple_model) %>% mutate(model = "Multiple: BMI + BP + S5")
) %>%
  select(model, r.squared, adj.r.squared, sigma)

print(model_comparison)
```

```
## # A tibble: 2 × 4
##   model                   r.squared adj.r.squared sigma
##   <chr>                       <dbl>         <dbl> <dbl>
## 1 Simple: BMI                 0.344         0.342  62.5
## 2 Multiple: BMI + BP + S5     0.459         0.457  56.8
```

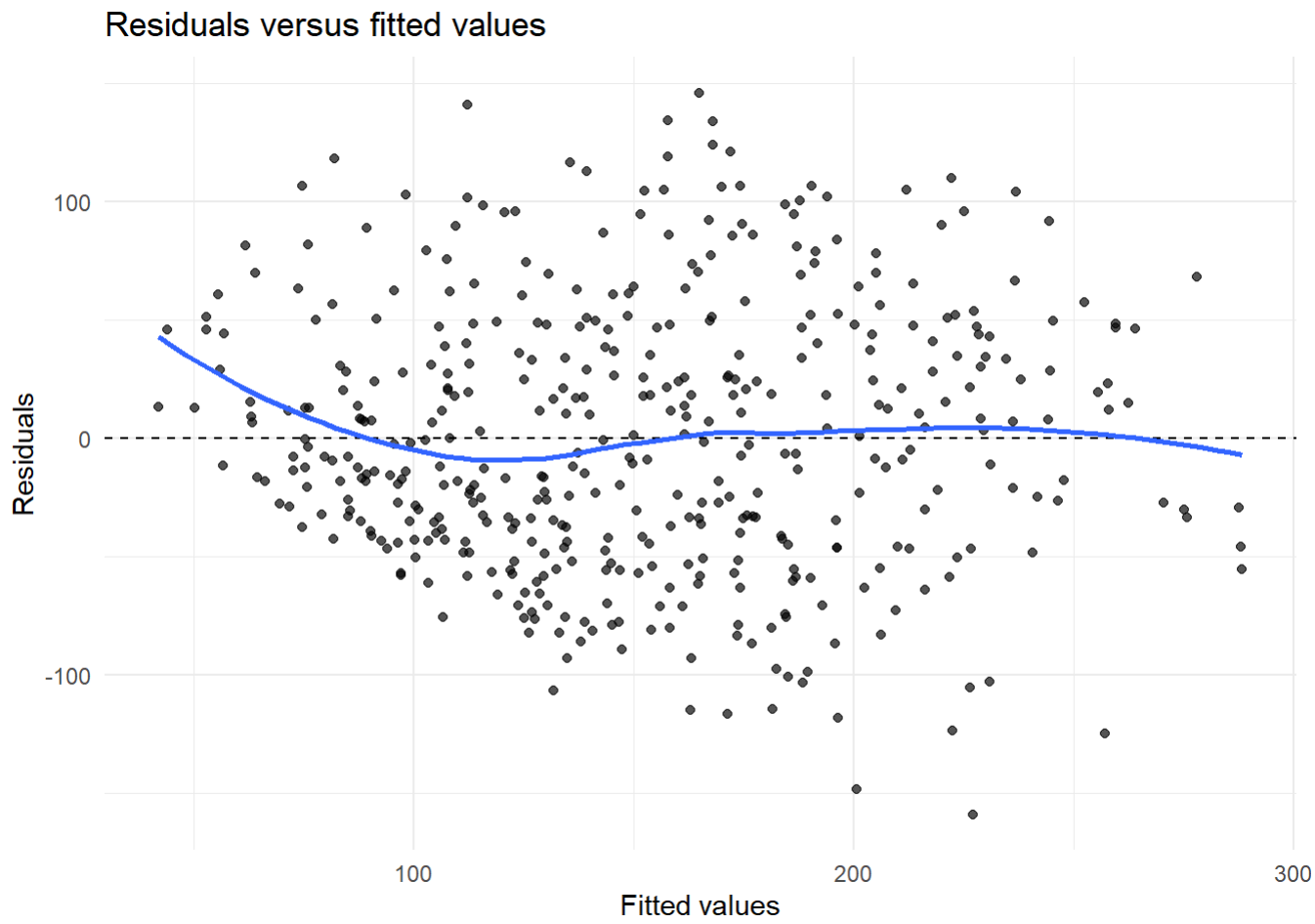
```r
# FOR YOUR REPORT
# - Interpret at least two coefficients from the multiple model.
# - Use the phrase 'holding the other predictors constant' correctly.
# - Compare the simple and multiple models.
# - Do not assume that the model with the larger R-squared is automatically
#   better in every respect.


# ============================================================================
# QUESTION 5. EVALUATE THE ANALYSIS
# ============================================================================
```

```r
# augment() creates fitted values and residuals in a convenient data frame.
diagnostic_data <- augment(multiple_model)

ggplot(diagnostic_data, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.65) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_smooth(se = FALSE) +
  labs(
    title = "Residuals versus fitted values",
    x = "Fitted values",
    y = "Residuals"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
# Look for obvious structure rather than trying to prove that every assumption
# is perfectly satisfied. A strong curve or a clear change in spread would be
# warning signs for a simple linear model.
```

```
# VIF can be calculated without installing another package.
manual_vif <- function(model) {
  X <- model.matrix(model)[, -1, drop = FALSE]  # remove intercept

  vif_values <- sapply(seq_len(ncol(X)), function(i) {
    target <- X[, i]
    others <- X[, -i, drop = FALSE]

    if (ncol(others) == 0) {
      return(1)
    }

    r2 <- summary(lm(target ~ others))$r.squared
    1 / (1 - r2)
  })

  setNames(vif_values, colnames(X))
}

manual_vif(multiple_model)
```

```
##   bmi    s5
## 1.249 1.249
```

```
# There is no single universal VIF cut-off. Interpret the values together with
# the correlation matrix and with the purpose of the model.
```

```
# A tidy table with estimates, standard errors, t-values, p-values and 95%
# confidence intervals.
coef_table <- tidy(multiple_model, conf.int = TRUE)
print(coef_table)
```

```
## # A tibble: 3 × 7
##   term        estimate std.error statistic  p.value conf.low conf.high
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 (Intercept)  -300.      24.9      -12.0  5.25e-29  -349.     -251.
## 2 bmi             7.28     0.684     10.6  1.17e-23     5.93      8.62
## 3 s5             56.1      5.79       9.69 3.04e-20    44.7      67.4
```

```
# You can also obtain the confidence intervals directly.
confint(multiple_model, level = 0.95)
```

```
##                2.5 %   97.5 %
## (Intercept) -348.964 -250.952
## bmi            5.931    8.621
## s5            44.684   67.428
```

```
# FOR YOUR REPORT
# - Comment on the residuals-versus-fitted plot.
# - Use VIF and the correlation matrix to discuss multicollinearity.
# - For at least two important coefficients, discuss the estimate and either
#   the confidence interval, t-value or p-value.
# - Distinguish statistical significance from practical importance.
# - State the main statistical weaknesses or uncertainties in the analysis.


# ============================================================================
# QUESTION 6. DRAW APPROPRIATE CONCLUSIONS
# ============================================================================

# No new code is required for this question.
# Use the results you have already produced and think critically about what they
# do and do not support.

# FOR YOUR REPORT
# Consider the following points:
# - Is this dataset experimental or observational?
# - Can the regression coefficients be interpreted causally?
# - Could there be confounding variables that are not included in the dataset?
# - Could measurement issues affect the results?
# - Would the results necessarily generalise to another population?
# - Which conclusions are supported by the evidence?
# - Which claims would go beyond the evidence?


# ============================================================================
# SEPARATE DELIVERABLE: DECISION-MAKER PRESENTATION
# ============================================================================

# The presentation is NOT another analysis section and does not require new R
# code. It should communicate the most important results from the report to a
# health-service decision-maker who is not a statistician.
#
# Suggested 4 to 6 slide structure:
# 1. The question and the data
# 2. The most important patterns in the data
# 3. The main regression findings
# 4. What the results mean in practical terms
# 5. The main uncertainty or limitation
# 6. What should and should not be concluded
#
# Use only a small number of well-chosen figures or statistics. Avoid raw R
# output and unnecessary technical notation.

##############################################################################
# END OF STARTER SCRIPT
##############################################################################
```